PhosWatch — UML Design Documentation

Real-Time Equipment Monitoring System with AI/ML

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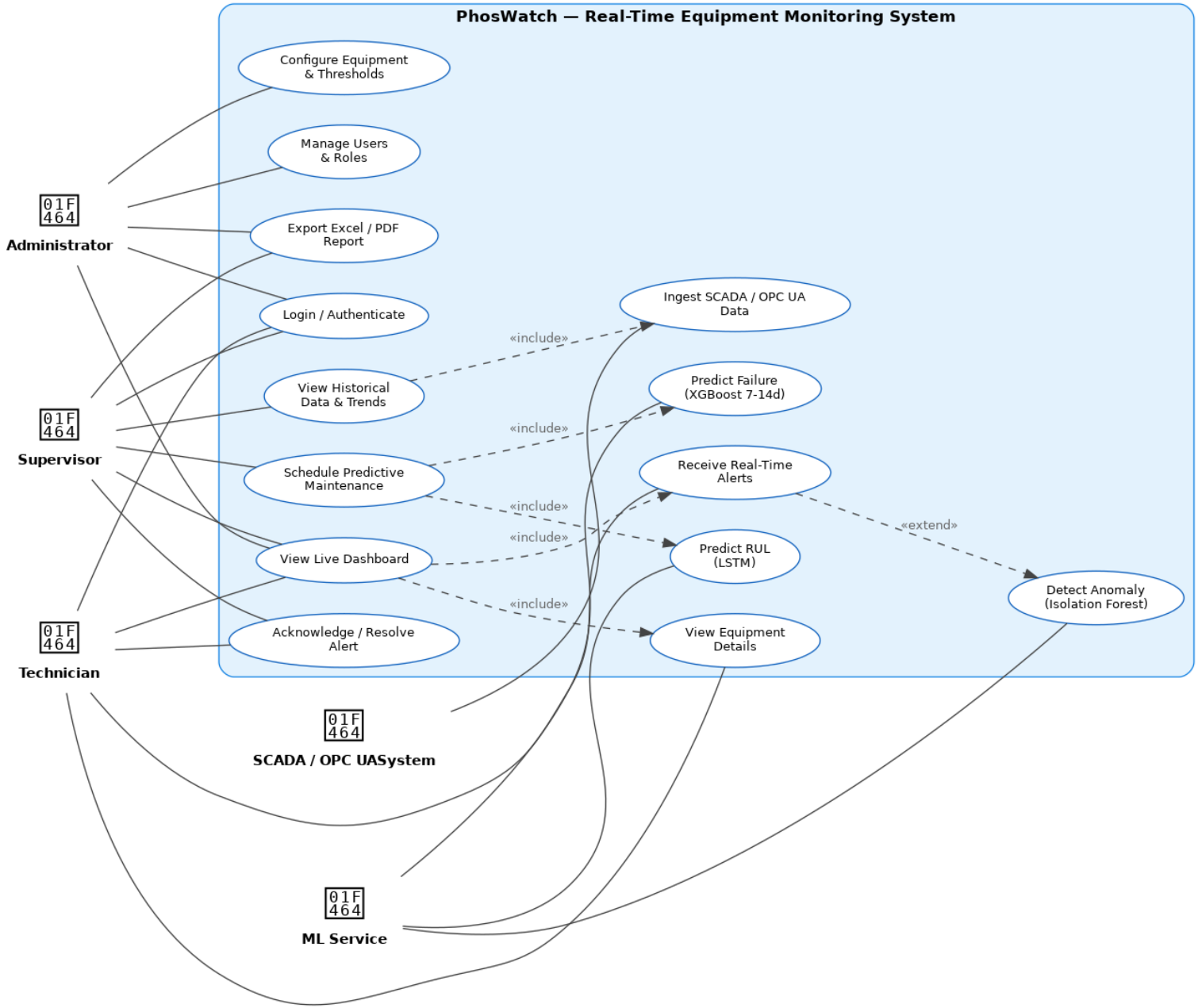
Abstract. This document presents the three core UML views of the PhosWatch monitoring platform: the **class diagram** (static structure of the domain and services), the **use case diagram** (system functionalities per actor), and the **sequence diagram** (runtime interaction for real-time monitoring, anomaly detection and predictive maintenance). The architecture follows a Docker microservices approach with four services: Backend (Node.js/Express), ML Service (Python/FastAPI with Isolation Forest, XGBoost and LSTM), Frontend (React/Recharts) and Database (PostgreSQL + TimescaleDB).

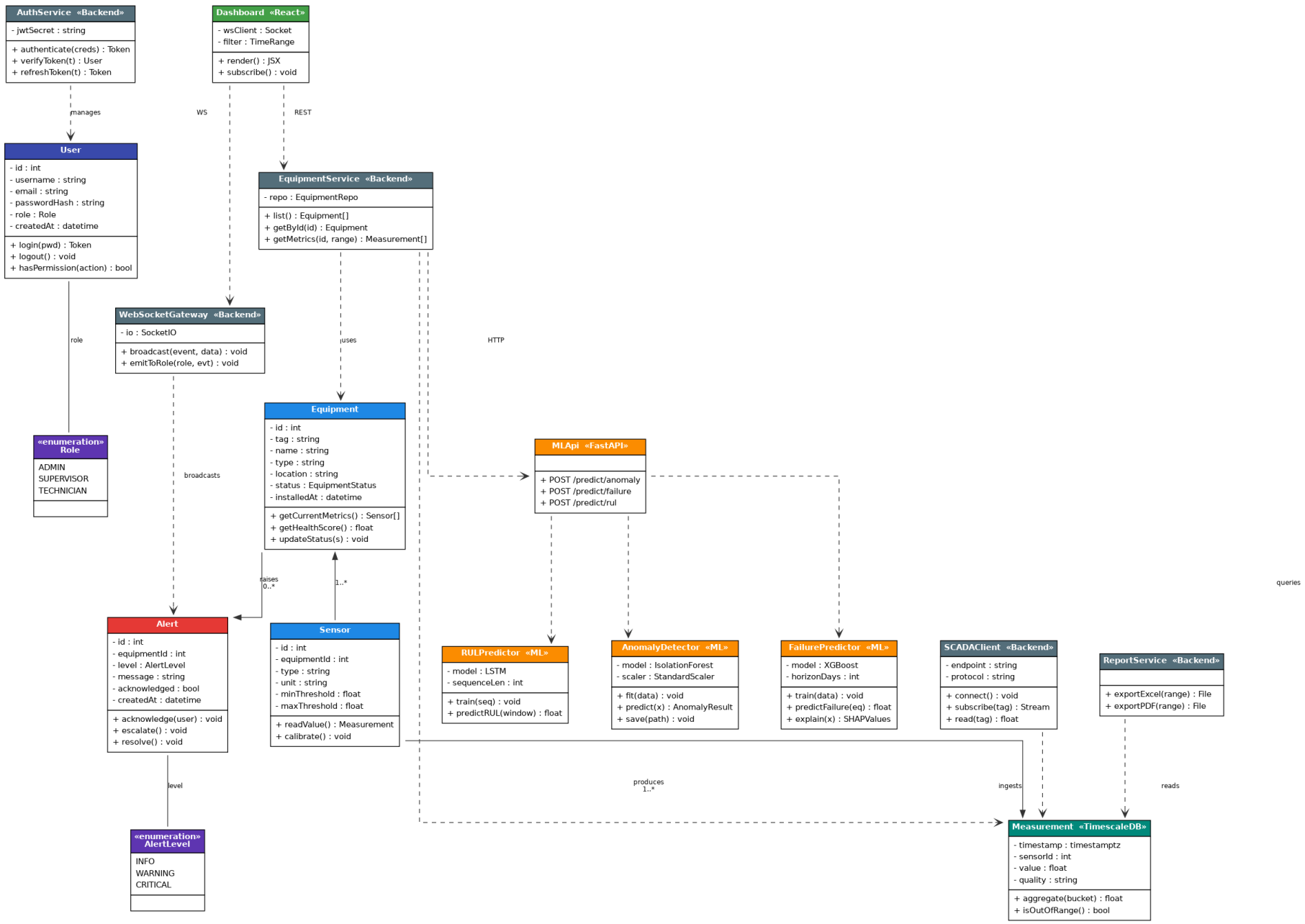
1. Use Case Diagram

Functional view — Actors (Admin, Supervisor, Technician, SCADA, ML Service) and the system's use cases

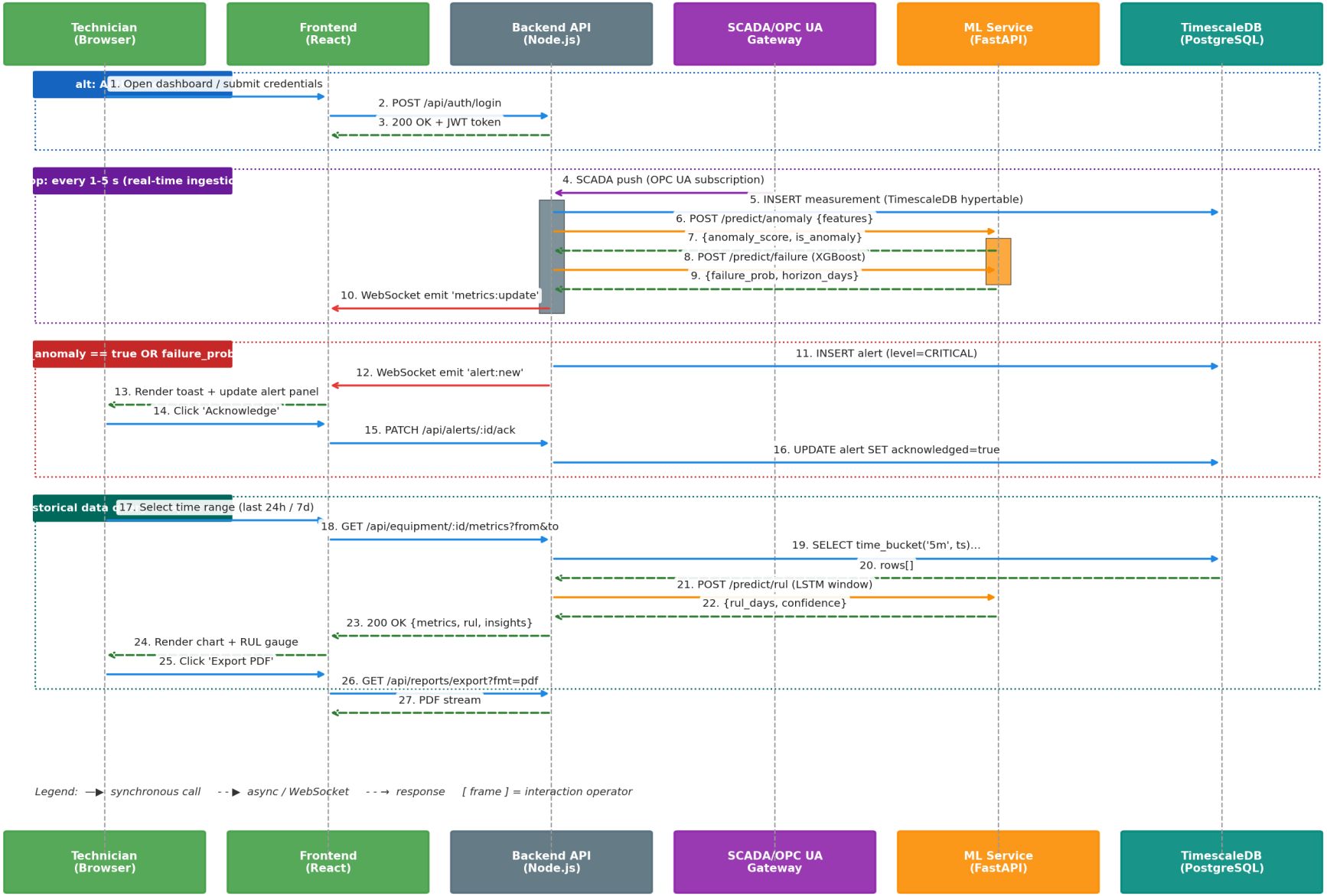

PhosWatch — Real-Time Equipment Monitoring System
Configure Equipment & Thresholds
Manage Users & Roles
Export Excel / PDF Report
Login / Authenticate
View Historical Data & Trends
Schedule Predictive Maintenance
View Live Dashboard
Acknowledge / Resolve Alert
Ingest SCADA / OPC UA Data
Predict Failure (XGBoost 7-14d)
Receive Real-Time Alerts
Predict RUL (LSTM)
View Equipment Details
Detect Anomaly (Isolation Forest)
«include»
«include»
«include»
«include»
«include»
«extend»
01F 464
Administrator
01F 464
Supervisor
01F 464
Technician
01F 464
SCADA / OPC UASystem
01F 464
ML Service

2. Class Diagram

Static structural view — Domain entities, backend services, ML components and their relationships


AuthService «Backend»
- jwtSecret : string
+ authenticate(creds) : Token
+ verifyToken(t) : User
+ refreshToken(t) : Token
Dashboard «React»
- wsClient : Socket
- filter : TimeRange
+ render() : JSX
+ subscribe() : void
manages
WS
REST
User
- id : int
- username : string
- email : string
- passwordHash : string
- role : Role
- createdAt : datetime
+ login(pwd) : Token
+ logout() : void
+ hasPermission(action) : bool
EquipmentService «Backend»
- repo : EquipmentRepo
+ list() : Equipment[]
+ getById(id) : Equipment
+ getMetrics(id, range) : Measurement[]
WebSocketGateway «Backend»
- io : SocketIO
+ broadcast(event, data) : void
+ emitToRole(role, evt) : void
role
uses
HTTP
«enumeration» Role
ADMIN
SUPERVISOR
TECHNICIAN
broadcasts
Equipment
- id : int
- tag : string
- name : string
- type : string
- location : string
- status : EquipmentStatus
- installedAt : datetime
+ getCurrentMetrics() : Sensor[]
+ getHealthScore() : float
+ updateStatus(s) : void
MLApi «FastAPI»
+ POST /predict/anomaly
+ POST /predict/failure
+ POST /predict/rul
raises
0..*
1..*
queries
Alert
- id : int
- equipmentId : int
- level : AlertLevel
- message : string
- acknowledged : bool
- createdAt : datetime
+ acknowledge(user) : void
+ escalate() : void
+ resolve() : void
Sensor
- id : int
- equipmentId : int
- type : string
- unit : string
- minThreshold : float
- maxThreshold : float
+ readValue() : Measurement
+ calibrate() : void
RULPredictor «ML»
- model : LSTM
- sequenceLen : int
+ train(seq) : void
+ predictRUL(window) : float
AnomalyDetector «ML»
- model : IsolationForest
- scaler : StandardScaler
+ fit(data) : void
+ predict(x) : AnomalyResult
+ save(path) : void
FailurePredictor «ML»
- model : XGBoost
- horizonDays : int
+ train(data) : void
+ predictFailure(eq) : float
+ explain(x) : SHAPValues
SCADAClient «Backend»
- endpoint : string
- protocol : string
+ connect() : void
+ subscribe(tag) : Stream
+ read(tag) : float
ReportService «Backend»
+ exportExcel(range) : File
+ exportPDF(range) : File
level
produces
1..*
ingests
reads
«enumeration» AlertLevel
INFO
WARNING
CRITICAL
Measurement «TimescaleDB»
- timestamp : timestamptz
- sensorId : int
- value : float
- quality : string
+ aggregate(bucket) : float
+ isOutOfRange() : bool

3. Sequence Diagram

Dynamic view — End-to-end real-time flow: SCADA ingestion → anomaly detection → WebSocket alert → acknowledgement → historical query & PDF export

Sequence Diagram — Real-Time Monitoring & Predictive Alert Flow